Web Programming Lab - Basic Solutions

Experiments 1 to 20 (parts a and b). Each program is kept as simple as possible.

Setup (for 3b, 4b, 6b)

Run these MySQL commands once to create the sample database.

```
CREATE DATABASE supermarket;
USE supermarket;
CREATE TABLE products (id INT PRIMARY KEY, name VARCHAR(50), brand VARCHAR(50), price DECIMAL(8,2));
INSERT INTO products VALUES
 (1,'Rice','Aashirvaad',60), (2,'Milk','Amul',25), (3,'Biscuit','Britannia',20),
 (4,'Soap','Dove',45), (5,'Tea','Tata',120);

-- Then in the project folder:  npm install mysql2
-- Run with:  node filename.js
```

Experiment 1

(a) Aim: To create a web page using all the basic HTML elements.

```
<!DOCTYPE html>
<html>
<head><title>Basic HTML Elements</title></head>
<body>
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<p>This is a <b>bold</b>, <i>italic</i> and <u>underlined</u> text.</p>
<hr>
<a href="https://www.google.com">Link</a><br>
<img src="photo.jpg" alt="Image" width="100"><br>
<ul><li>Item 1</li><li>Item 2</li></ul>
<ol><li>First</li><li>Second</li></ol>
<table border="1">
  <tr><th>Name</th><th>Age</th></tr>
  <tr><td>Ram</td><td>20</td></tr>
</table>
<form>
  Name: <input type="text"><br>
  <input type="radio" name="g"> Male
  <input type="radio" name="g"> Female<br>
  <input type="checkbox"> Agree<br>
  <select><option>One</option><option>Two</option></select><br>
  <textarea rows="2" cols="20"></textarea><br>
  <button>Submit</button>
</form>
</body>
</html>
```

(b) Aim: To use built-in Angular directives (ng-show, ng-repeat) to show/hide elements and display a list.

```
<!DOCTYPE html>
<html>
<head>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</head>
<body ng-app="" ng-init="show=true; fruits=['Apple','Banana','Mango']">
  <button ng-click="show=!show">Show / Hide</button>
  <div ng-show="show">
    <h3>Fruit List</h3>
    <ul><li ng-repeat="f in fruits">{{ f }}</li></ul>
  </div>
</body>
</html>
```

Experiment 2

(a) Aim: To create a web page using inline, internal and external CSS with different selectors.

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="style.css">      <!-- External CSS -->
  <style>                                       /* Internal CSS */
    h2 { color: green; }
  </style>
</head>
<body>
  <h1>Heading (element selector, external)</h1>
  <h2>Heading 2 (internal CSS)</h2>
  <p class="red">Class selector</p>
  <p id="main">ID selector</p>
  <p style="color:purple;">Inline CSS</p>
  <div><p>Descendant selector</p></div>
  <a href="#">Hover over me</a>
</body>
</html>

/* style.css */
h1 { text-align: center; }
.red { color: red; }
#main { background-color: yellow; }
div p { font-size: 20px; }
a:hover { color: green; }
```

(b) Aim: To design a shopping cart using AngularJS (select items by category, submit, show cart with price).

```
<!DOCTYPE html>
<html>
<head>
<script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
<script>
var app = angular.module('shop', []);
app.controller('cartCtrl', function($scope) {
  $scope.categories = {
    'Fruits':     [{name:'Apple', price:50},  {name:'Banana', price:30}],
    'Vegetables': [{name:'Tomato', price:20}, {name:'Potato', price:25}],
    'Dairy':      [{name:'Milk', price:40},   {name:'Curd', price:35}]
  };
  $scope.category = 'Fruits';
  $scope.cart = [];
  $scope.total = 0;
  $scope.submit = function() {
    $scope.cart = [];
    $scope.total = 0;
    angular.forEach($scope.categories, function(items) {
      angular.forEach(items, function(i) {
        if (i.selected) { $scope.cart.push(i); $scope.total += i.price; }
      });
    });
  };
});
</script>
</head>
<body ng-app="shop" ng-controller="cartCtrl">
  <h2>Shopping Cart</h2>
  Category:
  <select ng-model="category"
          ng-options="key as key for (key, value) in categories"></select>
  <div ng-repeat="i in categories[category]">
    <input type="checkbox" ng-model="i.selected"> {{i.name}} - Rs.{{i.price}}
  </div>
  <br><button ng-click="submit()">Submit</button>

  <h3>Your Cart</h3>
  <table border="1">
    <tr><th>Item</th><th>Price</th></tr>
    <tr ng-repeat="c in cart"><td>{{c.name}}</td><td>{{c.price}}</td></tr>
  </table>
  <h3>Total: Rs.{{total}}</h3>
</body>
```

```
</html>
```

Experiment 3

(a) Aim: To create an image slider with Previous / Next buttons to move between images manually.

```
<!DOCTYPE html>
<html>
<body>
  <img id="slide" src="img1.jpg" width="400" height="250"><br>
  <button onclick="change(-1)">Previous</button>
  <button onclick="change(1)">Next</button>
<script>
var images = ["img1.jpg", "img2.jpg", "img3.jpg"];
var i = 0;
function change(n) {
  i = (i + n + images.length) % images.length;
  document.getElementById("slide").src = images[i];
}
</script>
</body>
</html>
```

(b) Aim: To create a Node.js + MySQL command line application that searches a product by ID or name.

```
// search.js
const mysql = require('mysql2');
const readline = require('readline');

const db = mysql.createConnection({
  host: 'localhost', user: 'root', password: 'root', database: 'supermarket'
});
const rl = readline.createInterface({ input: process.stdin, output: process.stdout });

rl.question('Enter product ID or name: ', (text) => {
  const sql = 'SELECT * FROM products WHERE id = ? OR name LIKE ?';
  db.query(sql, [text, '%' + text + '%'], (err, rows) => {
    if (err) throw err;
    if (rows.length == 0) console.log('Product not found');
    else console.table(rows);
    db.end();
    rl.close();
  });
});
```

Experiment 4

(a) Aim: To create an automatic image slider that changes images by itself.

```
<!DOCTYPE html>
<html>
<body>
  <img id="slide" src="img1.jpg" width="400" height="250">
<script>
var images = ["img1.jpg", "img2.jpg", "img3.jpg"];
var i = 0;
setInterval(function() {
  i = (i + 1) % images.length;
  document.getElementById("slide").src = images[i];
}, 2000);      // change every 2 seconds
</script>
</body>
</html>
```

(b) Aim: To create a Node.js + MySQL command line application that searches a product by name.

```
// searchname.js
const mysql = require('mysql2');
const readline = require('readline');

const db = mysql.createConnection({
  host: 'localhost', user: 'root', password: 'root', database: 'supermarket'
});
const rl = readline.createInterface({ input: process.stdin, output: process.stdout });

rl.question('Enter product name: ', (name) => {
  db.query('SELECT * FROM products WHERE name LIKE ?', ['%' + name + '%'], (err, rows) => {
    if (err) throw err;
    if (rows.length == 0) console.log('Product not found');
    else console.table(rows);
    db.end();
    rl.close();
  });
});
```

Experiment 5

(a) Aim: To experiment with margin, padding and border (box model) using inline and external CSS.

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="box.css">
</head>
<body>
  <div class="box1">External CSS box</div>
  <div style="width:200px; height:60px; margin:20px; padding:15px;
              border:3px dashed red; background-color:lightyellow;">
    Inline CSS box
  </div>
</body>
</html>

/* box.css */
.box1 {
  width: 250px;
  height: 80px;
  margin: 30px;
  padding: 20px;
  border: 5px solid blue;
  background-color: lightblue;
}
```

(b) Aim: To write a JavaScript program for FizzBuzz (1 to 100).

```
// fizzbuzz.js     (run: node fizzbuzz.js)
for (let i = 1; i <= 100; i++) {
  if (i % 15 == 0) console.log("FizzBuzz");
  else if (i % 3 == 0) console.log("Fizz");
  else if (i % 5 == 0) console.log("Buzz");
  else console.log(i);
}
```

Experiment 6

(a) Aim: To create a digital clock showing the current time.

```
<!DOCTYPE html>
<html>
<body>
  <h1 id="clock"></h1>
<script>
function showTime() {
  document.getElementById("clock").innerHTML = new Date().toLocaleTimeString();
}
showTime();
setInterval(showTime, 1000);
</script>
</body>
</html>
```

(b) Aim: To display product details of different brands in table format with the Price sorted (Node.js + MySQL).

```
// sorted.js
const mysql = require('mysql2');

const db = mysql.createConnection({
  host: 'localhost', user: 'root', password: 'root', database: 'supermarket'
});

db.query('SELECT id, name, brand, price FROM products ORDER BY price ASC', (err, rows) => {
  if (err) throw err;
  console.table(rows);
  db.end();
});
```

Experiment 7

(a) Aim: To create a digital clock showing the current time and date.

```
<!DOCTYPE html>
<html>
<body>
  <h2 id="date"></h2>
  <h1 id="time"></h1>
<script>
function showClock() {
  var d = new Date();
  document.getElementById("date").innerHTML = d.toLocaleDateString();
  document.getElementById("time").innerHTML = d.toLocaleTimeString();
}
showClock();
setInterval(showClock, 1000);
</script>
</body>
</html>
```

(b) Aim: To build a web app that fetches a random quote from an API when a button is clicked.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="getQuote()">New Quote</button>
  <p id="quote"></p>
  <p id="author"></p>
<script>
function getQuote() {
  fetch("https://dummyjson.com/quotes/random")
    .then(function(r) { return r.json(); })
    .then(function(d) {
      document.getElementById("quote").innerHTML = d.quote;
      document.getElementById("author").innerHTML = "- " + d.author;
    });
}
</script>
</body>
</html>
```

Experiment 8

(a) Aim: To style a basic HTML page using CSS (fonts, colors, margins, padding).

```
<!DOCTYPE html>
<html>
<head>
<style>
  body { font-family: Arial; background-color: #f0f0f0; margin: 20px; }
  h1 { color: darkblue; font-size: 32px; }
  p { color: #333; font-size: 18px; padding: 10px; margin: 15px; }
</style>
</head>
<body>
  <h1>My Page</h1>
  <p>This is a styled paragraph.</p>
</body>
</html>
```

(b) Aim: To create a JavaScript function that finds the sum of all prime numbers up to a given limit.

```
// primesum.js
function sumPrimes(limit) {
  let sum = 0;
  for (let n = 2; n <= limit; n++) {
    let prime = true;
    for (let i = 2; i * i <= n; i++) {
      if (n % i == 0) { prime = false; break; }
    }
    if (prime) sum += n;
  }
  return sum;
}
console.log(sumPrimes(20));     // Output: 77
```

Experiment 9

(a) Aim: To write client-side scripts for validating web form controls using DHTML.

```
<!DOCTYPE html>
<html>
<body>
<form onsubmit="return validate()">
  Name: <input type="text" id="name"><br>
  Email: <input type="text" id="email"><br>
  Password: <input type="password" id="pass"><br>
  <input type="submit" value="Submit">
</form>
<p id="msg" style="color:red"></p>
<script>
function validate() {
  var n = document.getElementById("name").value;
  var e = document.getElementById("email").value;
  var p = document.getElementById("pass").value;
  var msg = document.getElementById("msg");
  if (n == "") { msg.innerHTML = "Name is required"; return false; }
  if (e.indexOf("@") == -1) { msg.innerHTML = "Enter a valid email"; return false; }
  if (p.length < 6) { msg.innerHTML = "Password must be 6+ characters"; return false; }
  alert("Form is valid");
  return true;
}
</script>
</body>
</html>
```

(b) Aim: To create a JavaScript function that finds whether a number is odd or even.

```
// oddeven.js
function oddEven(n) {
  if (n % 2 == 0) return n + " is even";
  else return n + " is odd";
}
console.log(oddEven(7));        // Output: 7 is odd
console.log(oddEven(10));       // Output: 10 is even
```

Experiment 10

(a) Aim: To create a webpage with a button that shows an alert when clicked.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="alert('Button clicked!')">Click Me</button>
</body>
</html>
```

(b) Aim: To generate a random number between 1 and 10 and display it on the page.

```
<!DOCTYPE html>
<html>
<body>
  <p id="r"></p>
<script>
var n = Math.floor(Math.random() * 10) + 1;
document.getElementById("r").innerHTML = "Random number: " + n;
</script>
</body>
</html>
```

Experiment 11

(a) Aim: To build a quiz with multiple-choice questions and give feedback on correct / incorrect answers.

```
<!DOCTYPE html>
<html>
<body>
  <h2>Quiz</h2>
  <p>1. Capital of India?</p>
  <input type="radio" name="q1" value="Delhi"> Delhi
  <input type="radio" name="q1" value="Mumbai"> Mumbai
  <p>2. 2 + 2 = ?</p>
  <input type="radio" name="q2" value="4"> 4
  <input type="radio" name="q2" value="5"> 5
  <br><br>
  <button onclick="check()">Submit</button>
  <p id="result"></p>
<script>
function check() {
  var answers = { q1: "Delhi", q2: "4" };
  var score = 0, text = "";
  for (var q in answers) {
    var sel = document.querySelector('input[name="' + q + '"]:checked');
    if (sel && sel.value == answers[q]) { score++; text += q + ": Correct<br>"; }
    else text += q + ": Wrong (Answer: " + answers[q] + ")<br>";
  }
  document.getElementById("result").innerHTML = text + "Score: " + score + "/2";
}
</script>
</body>
</html>
```

(b) Aim: To write a JavaScript program to check whether a given year is a leap year.

```
// leap.js
function isLeap(y) {
  return (y % 4 == 0 && y % 100 != 0) || (y % 400 == 0);
}
var year = 2024;
if (isLeap(year)) console.log(year + " is a leap year");
else console.log(year + " is not a leap year");
```

Experiment 12

(a) Aim: To create a webpage with a button that changes the text on the page when clicked.

```
<!DOCTYPE html>
<html>
<body>
  <p id="text">Hello! This is the old text.</p>
  <button onclick="change()">Change Text</button>
<script>
function change() {
  document.getElementById("text").innerHTML = "The text has been changed!";
}
</script>
</body>
</html>
```

(b) Aim: To design a JavaScript function that calculates the square of a number and displays the result.

```
<!DOCTYPE html>
<html>
<body>
  <p id="out"></p>
<script>
function square(n) {
  return n * n;
}
document.getElementById("out").innerHTML = "Square of 5 is " + square(5);
</script>
</body>
</html>
```

Experiment 13

(a) Aim: To build a form validation script that checks all required fields are filled before submission.

```
<!DOCTYPE html>
<html>
<body>
<form onsubmit="return check()">
  Name: <input type="text" id="n"><br>
  Email: <input type="text" id="e"><br>
  Phone: <input type="text" id="p"><br>
  <input type="submit" value="Submit">
</form>
<script>
function check() {
  var ids = ["n", "e", "p"];
  for (var i = 0; i < ids.length; i++) {
    if (document.getElementById(ids[i]).value == "") {
      alert("Please fill all the fields");
      return false;
    }
  }
  alert("Form submitted");
  return true;
}
</script>
</body>
</html>
```

(b) Aim: To create a JavaScript array of favourite fruits and display them one by one using a loop.

```
// fruits.js
var fruits = ["Apple", "Mango", "Banana", "Orange", "Grapes"];
for (var i = 0; i < fruits.length; i++) {
  console.log(fruits[i]);
}
```

Experiment 14

(a) Aim: To create a web page with images and embedded audio and video content.

```
<!DOCTYPE html>
<html>
<body>
  <h2>Image</h2>
  <img src="photo.jpg" alt="Photo" width="300">

  <h2>Audio</h2>
  <audio controls>
    <source src="song.mp3" type="audio/mpeg">
  </audio>

  <h2>Video</h2>
  <video width="320" height="240" controls>
    <source src="movie.mp4" type="video/mp4">
  </video>
</body>
</html>
```

(b) Aim: To create a JavaScript program that prompts the user for their name and displays a greeting.

```
<!DOCTYPE html>
<html>
<body>
<script>
var userName = prompt("Enter your name");
alert("Hello, " + userName + "!");
document.write("Hello, " + userName + "! Welcome.");
</script>
</body>
</html>
```

Experiment 15

(a) Aim: To style text (color, font size, font family) and apply background colors/images using CSS selectors.

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Heading</h1>
  <p class="note">Paragraph with a class.</p>
  <div id="box">Div with a background image</div>
</body>
</html>

/* style.css */
body { background-color: lightyellow; }
h1 { color: red; font-size: 36px; font-family: Verdana; }
.note { color: blue; font-size: 20px; font-family: "Times New Roman"; }
#box {
  height: 100px;
  color: white;
  background-image: url("bg.jpg");
  background-size: cover;
}
```

(b) Aim: To enhance a calculator with square root and exponentiation using JavaScript.

```
<!DOCTYPE html>
<html>
<body>
  Number 1: <input type="number" id="a"><br>
  Number 2: <input type="number" id="b"><br><br>
  <button onclick="calc('+')">+</button>
  <button onclick="calc('-')">-</button>
  <button onclick="calc('*')">*</button>
  <button onclick="calc('/')">/</button>
  <button onclick="calc('sqrt')">Square Root (of No.1)</button>
  <button onclick="calc('pow')">Power (No.1 ^ No.2)</button>
  <h3 id="res"></h3>
<script>
function calc(op) {
  var a = Number(document.getElementById("a").value);
  var b = Number(document.getElementById("b").value);
  var r;
  if (op == "+") r = a + b;
  else if (op == "-") r = a - b;
  else if (op == "*") r = a * b;
  else if (op == "/") r = a / b;
  else if (op == "sqrt") r = Math.sqrt(a);
  else if (op == "pow") r = Math.pow(a, b);
  document.getElementById("res").innerHTML = "Result: " + r;
}
</script>
</body>
</html>
```

Experiment 16

(a) Aim: To experiment with margin, padding and border properties and their effect on the box model.

```
<!DOCTYPE html>
<html>
<head>
<style>
  div { background-color: lightgreen; }
  .a { width: 200px; height: 60px; margin: 10px; padding: 10px; border: 2px solid black; }
  .b { width: 300px; height: 100px; margin: 30px; padding: 25px; border: 5px dotted red; }
</style>
</head>
<body>
  <div class="a">Box A: small margin and padding</div>
  <div class="b">Box B: big margin and padding</div>
</body>
</html>
```

(b) Aim: To build a currency converter using JavaScript (rates below are sample values).

```
<!DOCTYPE html>
<html>
<body>
  Amount: <input type="number" id="amt">
  <select id="cur">
    <option value="0.012">INR to USD</option>
    <option value="0.011">INR to EUR</option>
    <option value="83">USD to INR</option>
    <option value="90">EUR to INR</option>
  </select>
  <button onclick="convert()">Convert</button>
  <h3 id="out"></h3>
<script>
function convert() {
  var amount = Number(document.getElementById("amt").value);
  var rate = Number(document.getElementById("cur").value);
  document.getElementById("out").innerHTML = "Converted: " + (amount * rate).toFixed(2);
}
</script>
</body>
</html>
```

Experiment 17

(a) Aim: To enhance a form with JavaScript validation (required fields, valid email, password strength).

```
<!DOCTYPE html>
<html>
<body>
<form onsubmit="return validate()">
  Name: <input type="text" id="name"><br>
  Email: <input type="text" id="email"><br>
  Password: <input type="password" id="pass"><br>
  <input type="submit" value="Register">
</form>
<script>
function validate() {
  var name = document.getElementById("name").value;
  var email = document.getElementById("email").value;
  var pass = document.getElementById("pass").value;
  var emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,}$/;
  var strong = /^(?=.*[A-Z])(?=.*[0-9])(?=.*[!@#$%^&*]).{8,}$/;
  if (name == "" || email == "" || pass == "") {
    alert("All fields are required"); return false;
  }
  if (!emailPattern.test(email)) { alert("Invalid email"); return false; }
  if (!strong.test(pass)) {
    alert("Password needs 8+ chars, a capital letter, a number and a symbol");
    return false;
  }
  alert("Registered successfully");
  return true;
}
</script>
</body>
</html>
```

(b) Aim: To build a simple calculator using HTML buttons and JavaScript for basic arithmetic.

```
<!DOCTYPE html>
<html>
<body>
  <input type="text" id="d" readonly><br>
  <button onclick="add('1')">1</button>
  <button onclick="add('2')">2</button>
  <button onclick="add('3')">3</button>
  <button onclick="add('+')">+</button><br>
  <button onclick="add('4')">4</button>
  <button onclick="add('5')">5</button>
  <button onclick="add('6')">6</button>
  <button onclick="add('-')">-</button><br>
  <button onclick="add('7')">7</button>
  <button onclick="add('8')">8</button>
  <button onclick="add('9')">9</button>
  <button onclick="add('*')">*</button><br>
  <button onclick="add('0')">0</button>
  <button onclick="clr()">C</button>
  <button onclick="equal()">=</button>
  <button onclick="add('/')">/</button>
<script>
function add(v) { document.getElementById("d").value += v; }
function clr() { document.getElementById("d").value = ""; }
function equal() {
  var d = document.getElementById("d");
  d.value = eval(d.value);
}
</script>
</body>
</html>
```

Experiment 18

(a) Aim: To create a password checker that checks length, numbers and special characters.

```
<!DOCTYPE html>
<html>
<body>
<script>
var pwd = prompt("Enter a password");
var msg = "";
if (pwd.length < 8) msg += "Minimum 8 characters. ";
if (!/[0-9]/.test(pwd)) msg += "Add a number. ";
if (!/[!@#$%^&*]/.test(pwd)) msg += "Add a special character. ";

if (msg == "") alert("Strong password");
else alert("Weak password: " + msg);
</script>
</body>
</html>
```

(b) Aim: To create a to-do list application to add, remove and mark tasks as complete.

```
<!DOCTYPE html>
<html>
<body>
  <h2>To-Do List</h2>
  <input type="text" id="task">
  <button onclick="addTask()">Add</button>
  <ul id="list"></ul>
<script>
function addTask() {
  var t = document.getElementById("task").value;
  if (t == "") return;
  var li = document.createElement("li");
  li.innerHTML = '<input type="checkbox" onchange="done(this)"> ' +
                 '<span>' + t + '</span> ' +
                 '<button onclick="this.parentNode.remove()">Remove</button>';
  document.getElementById("list").appendChild(li);
  document.getElementById("task").value = "";
}
function done(box) {
  box.nextElementSibling.style.textDecoration = box.checked ? "line-through" : "none";
}
</script>
</body>
</html>
```

Experiment 19

(a) Aim: To apply internal and external CSS to HTML elements to change colors, fonts and backgrounds.

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="style.css">     <!-- External CSS -->
  <style>                                      /* Internal CSS */
    p { color: darkgreen; font-size: 20px; background-color: #e0ffe0; }
  </style>
</head>
<body>
  <h1>External CSS heading</h1>
  <p>Internal CSS paragraph</p>
</body>
</html>

/* style.css */
body { background-color: lightyellow; }
h1 { color: red; font-family: Verdana; font-size: 30px; }
```

(b) Aim: To find the factorial of a number (taken from a prompt) and display it on clicking a Submit button.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="fact()">Submit</button>
  <p id="out"></p>
<script>
function fact() {
  var n = Number(prompt("Enter a number"));
  var f = 1;
  for (var i = 1; i <= n; i++) {
    f = f * i;
  }
  document.getElementById("out").innerHTML = "Factorial of " + n + " = " + f;
}
</script>
</body>
</html>
```

Experiment 20

(a) Aim: To style form elements (inputs, buttons, radio buttons, checkboxes) using external and embedded CSS.

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="form.css">      <!-- External CSS -->
  <style>                                      /* Embedded CSS */
    button { background-color: green; color: white; padding: 8px 20px; border: none; }
  </style>
</head>
<body>
<form>
  Name: <input type="text"><br><br>
  Gender:
  <input type="radio" name="g"> Male
  <input type="radio" name="g"> Female<br><br>
  <input type="checkbox"> I agree<br><br>
  <button type="button">Submit</button>
</form>
</body>
</html>

/* form.css */
input[type="text"] { width: 200px; padding: 6px; border: 2px solid blue; border-radius: 5px; }
input[type="radio"], input[type="checkbox"] { width: 18px; height: 18px; accent-color: red; }
```

(b) Aim: To find the Fibonacci series of a number (taken from a prompt) and display it on clicking a Submit button.

```
<!DOCTYPE html>
<html>
<body>
  <button onclick="fib()">Submit</button>
  <p id="out"></p>
<script>
function fib() {
  var n = Number(prompt("How many terms?"));
  var a = 0, b = 1, series = "";
  for (var i = 1; i <= n; i++) {
    series += a + " ";
    var next = a + b;
    a = b;
    b = next;
  }
  document.getElementById("out").innerHTML = "Fibonacci: " + series;
}
</script>
</body>
</html>
```